CoLoRe installation instructions for NERSC

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1 Create conda environment

Create an empty conda environment and install the following C libraries from the `conda-forge` channel: `libconfig, libsharp, cfitsio, healpix`.
You can do so with the following commands:

```
conda create -n colore_env
conda activate colore_env
conda install -c conda-forge libconfig
conda install -c conda-forge libsharp
conda install -c conda-forge cfitsio
conda install -c conda-forge healpix
```

2 Environmental variables

Make sure that the libraries used to compile CoLoRe are in `LD_LIBRARY_PATH` so they can be found at runtime. Open your `.bashrc` file and make the directories point to your new environment where the libraries have been installed.

```
export FFTW_DIR=/opt/cray/pe/fftw/3.3.10.11/x86_milan
export CFITSIO_DIR=/global/common/software/desi/user/.conda/envs/colore_env
export HEALPIX_DIR=/global/common/software/desi/user/.conda/envs/colore_env
export LIBSHARP_DIR=/global/common/software/desi/user/.conda/envs/colore_env
export LIBCONFIG_DIR=/global/common/software/desi/user/.conda/envs/colore_env

export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:/usr/lib64:${FFTW_DIR}/lib:${CFITSIO_DIR}/lib:${LIBSHARP_DIR}/a
↪  uto/lib:${LIBCONFIG_DIR}/lib
```

3 Modify CoLoRe's makefile

Clone CoLoRe GitHub repo into your home directory.

```
git clone https://github.com/damonge/CoLoRe
```

Inside the repo folder `./CoLoRe`, modify the `Makefile` to use the environmental variables defined earlier in the `.bashrc` file. A few other variables have to be changed since NERSC has continued to update its modules and libraries. The compiler options for example needs the `-fpermissive` tag to allow for some differences between the compiler's NERSC and CoLoRe versions. Below you can find an example of the the modified `Makefile`.

```
########## User-definable stuff ##########
###Compiler and compilation options
COMP_SER = cc
COMP_MPI = cc
OPTIONS = -Wall -O3 -std=c99 -fpermissive


### Behavioural flags
#Use double precision integer (enable in general)
```

```
DEFINEFLAGS += -D_LONGIDS
#Use normalized bias model
DEFINEFLAGS += -D_BIAS_MODEL_2
#Use linear bias model
#DEFINEFLAGS += -D_BIAS_MODEL_3
#Use new lensing method
#DEFINEFLAGS += -D_USE_FAST_LENSING
#Generate debug help. Only useful for development
DEFINEFLAGS += -D_DEBUG
#Use double precision floating point? Set to "yes" or "no"
USE_SINGLE_PRECISION = yes
#Add random perturbations to kappa from redshifts outside the box
#ADD_EXTRA_KAPPA = no
#Compile with HDF5 capability? Set to "yes" or "no"
USE_HDF5 = no
#Use OMP parallelization? Set to "yes" or "no"
USE_OMP = yes
#Use MPI parallelization? Set to "yes" or "no"
USE_MPI = yes

###Path to libraries and headers
###If two or more of the dependencies reside in the same paths, only
###one instance is necessary.
#GSL
#GSL_INC = -I/add/path
#GSL_LIB = -L/add/path
GSL_INC = -I$/usr/include
GSL_LIB = -L/usr/lib64 -lgsl -lgslcblas

#FFTW
FFTW_INC = -I${FFTW_DIR}/include
FFTW_LIB = -L${FFTW_DIR}/lib

#cfitsio
FITS_INC = -I${CFITSIO_DIR}/include
FITS_LIB = -L${CFITSIO_DIR}/lib -lcfitsio

#cfitsio
HDF5_INC =
HDF5_LIB =

#libconfig
CONF_INC = -I${LIBCONFIG_DIR}/include
CONF_LIB = -L${LIBCONFIG_DIR}/lib

#healpix
HPIX_INC = -I${HEALPIX_DIR}/include
HPIX_LIB = -L${HEALPIX_DIR}/lib

#libsharp
SHT_INC = -I${LIBSHARP_DIR}/include
SHT_LIB = -L${LIBSHARP_DIR}/lib
```

This last part of the file remains the same.

4 Compile CoLoRe

Now everything is ready to compile the CoLoRe code. Go to the CoLoRe repo folder and use the `make` command inside it. To make sure it has compiled correctly, no error should appear as the last messages of the compilation.

Remember

Everytime CoLoRe is run from a script, the conda module has to be loaded and the environment has to be activated. You can just copy the following lines into your script.

```
module load conda
conda activate colore_env
```